

Final Project Report: Business Database Development

Course: Software Development / Database Systems

Project Title: Creating a Business Database from Scratch

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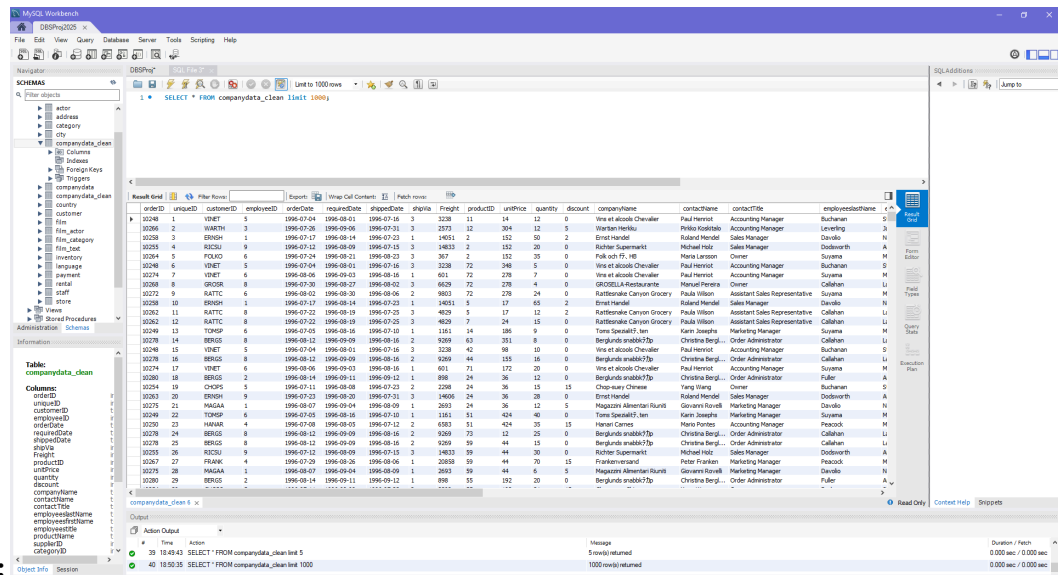
Introduction

This project involved the design and implementation of a fully functional business database using MySQL Workbench. The objective was to transform a raw data file into a normalized, relational database complete with views, triggers, stored procedures, user access controls, and a backup. The process was carried out in multiple phases, each focusing on a critical aspect of database development—from initial data cleaning to final system export.

Phase 1: Database Creation & Data Loading

The project began with a review of an Excel spreadsheet provided via Canvas. This file contained various business-related data that required cleaning before it could be loaded into a database. Date fields were reformatted into the standard yyyy-mm-dd format, and all currency symbols and numeric formatting such as commas were removed from monetary and numeric fields to ensure compatibility with SQL data types. Once cleaned, the data was saved as CompanyData_clean.csv.

A new database titled BusinessDB was then created in MySQL Workbench. Using the Table Import Wizard, the cleaned CSV file was successfully imported into the system, forming the foundation of the database.



The screenshot displays the MySQL Workbench interface. On the left, the 'Schemas' pane shows the 'companydata_clean' database selected. The 'Table' pane lists the columns of the 'companydata_clean' table: orderID, uniqueID, customerID, employeeID, orderDate, invoiceDate, shipperCode, shipVia, freight, productID, unitPrice, quantity, discount, companyName, contactName, contactTitle, employeeName, employeeID, employeeTitle, productID, productName, categoryID, categoryName, and categoryTitle. The 'Query' pane shows the SQL query: `SELECT * FROM companydata_clean limit 1000;`. The 'Result' pane displays the first 1000 rows of the query result, showing columns: orderID, uniqueID, customerID, employeeID, orderDate, invoiceDate, shipperCode, shipVia, freight, productID, unitPrice, quantity, discount, companyName, contactName, contactTitle, employeeName, employeeID, employeeTitle, productID, productName, categoryID, categoryName, and categoryTitle. The 'Output' pane shows the execution status: '39 10.49.35 SELECT * FROM companydata_clean limit 5' and '40 10.50.35 SELECT * FROM companydata_clean limit 1000'.

orderID	uniqueID	customerID	employeeID	orderDate	invoiceDate	shipperCode	shipVia	freight	productID	unitPrice	quantity	discount	companyName	contactName	contactTitle	employeeName	employeeID	employeeTitle	productID	productName	categoryID	categoryName	categoryTitle
10248	1	VENET	5	1996-07-04	1996-08-01	1996-07-06	3	3238	11	14	12	0	Vins et abricots Chevalier	Paul Henriot	Accounting Manager	Buchanan	5						
10266	2	WARTH	3	1996-07-26	1996-09-06	1996-07-31	3	2373	12	304	12	5	Warten Henkel	Polko Kozlowski	Accounting Manager	Lewelling	2						
10250	3	ZEMCH	1	1996-07-17	1996-08-14	1996-07-23	1	14951	2	152	50	2	Ernst Henkel	Rafael Henkel	Sales Manager	Davidis	9						
10255	4	RIZSU	9	1996-07-12	1996-08-20	1996-07-15	3	14833	2	152	20	0	Richter Supermarkt	Michael Holz	Sales Manager	Dodsworth	4						
10244	5	FOUCH	6	1996-07-24	1996-08-23	1996-08-23	3	267	2	152	20	0	Paulsen IT, HB	Mark Linton	Owner	Buchanan	9						
10246	6	VENET	5	1996-07-04	1996-08-01	1996-07-06	3	3238	72	248	5	0	Vins et abricots Chevalier	Paul Henriot	Accounting Manager	Buchanan	5						
10274	7	VENET	6	1996-08-06	1996-09-03	1996-08-10	1	601	72	238	7	0	Vins et abricots Chevalier	Paul Henriot	Accounting Manager	Sydney	9						
10268	8	GRUOS	8	1996-07-30	1996-08-27	1996-08-02	3	4629	22	238	4	0	GRUOSLA-Rochester	Renald Penne	Owner	Callahan	11						
10272	9	RATTC	8	1996-08-02	1996-09-30	1996-08-06	2	9803	72	238	24	0	Rattlesnake Canyon Grocery	Paul Wilson	Assistant Sales Representative	Sydney	9						
10258	10	BONSH	1	1996-07-17	1996-08-14	1996-07-23	1	14951	1	17	60	2	Ernst Henkel	Rafael Henkel	Sales Manager	Davidis	9						
10262	11	RATTC	8	1996-07-22	1996-08-19	1996-07-25	3	4629	5	17	12	2	Rattlesnake Canyon Grocery	Paul Wilson	Assistant Sales Representative	Callahan	11						
10262	12	RATTC	8	1996-07-22	1996-08-19	1996-07-25	3	4629	7	24	15	0	Rattlesnake Canyon Grocery	Paul Wilson	Assistant Sales Representative	Callahan	11						
10246	13	TONGP	6	1996-07-03	1996-08-18	1996-07-10	1	1161	14	186	9	0	Time-Sensitized, Inc.	Karen Joseph	Marketing Manager	Sydney	9						
10278	14	BERUS	8	1996-08-12	1996-09-09	1996-08-16	2	9269	63	331	8	0	Berglund's sm&bb7Pp	Christina Berglund	Order Administrator	Callahan	11						
10246	15	VENET	5	1996-07-04	1996-08-01	1996-07-06	3	3238	42	68	10	0	Vins et abricots Chevalier	Paul Henriot	Accounting Manager	Buchanan	5						
10278	16	BERUS	8	1996-08-12	1996-09-09	1996-08-16	2	9269	44	155	18	0	Berglund's sm&bb7Pp	Christina Berglund	Order Administrator	Callahan	11						
10274	17	VENET	6	1996-08-06	1996-09-03	1996-08-10	1	601	71	172	20	0	Vins et abricots Chevalier	Paul Henriot	Accounting Manager	Sydney	9						
10260	18	BERUS	2	1996-08-14	1996-09-11	1996-08-12	1	866	24	36	12	0	Berglund's sm&bb7Pp	Christina Berglund	Order Administrator	Fulmer	4						
10254	19	CHOPS	5	1996-07-11	1996-08-08	1996-07-23	2	2296	24	36	15	15	Chapman Cheese	Yang Wang	Owner	Buchanan	9						
10263	20	ERNDH	9	1996-07-23	1996-08-20	1996-07-31	3	14906	24	36	28	0	Ernst Henkel	Rafael Henkel	Sales Manager	Dodsworth	4						
10275	21	MAGAA	1	1996-08-07	1996-09-04	1996-08-09	1	2653	24	36	12	5	Maggiore Alimentari Runzi	Giovanni Rovelli	Marketing Manager	Davidis	9						
10246	22	TONGP	6	1996-07-03	1996-08-18	1996-07-10	1	1161	51	424	40	0	Time-Sensitized, Inc.	Karen Joseph	Marketing Manager	Sydney	9						
10250	23	HANAR	4	1996-07-08	1996-08-05	1996-07-12	2	6382	51	424	35	15	Hanan Carnes	Mario Pontes	Accounting Manager	Pearson	9						
10278	24	BERUS	8	1996-08-12	1996-09-09	1996-08-16	2	9269	75	12	25	0	Berglund's sm&bb7Pp	Christina Berglund	Order Administrator	Callahan	11						
10278	25	BERUS	8	1996-08-12	1996-09-09	1996-08-16	2	9269	59	44	15	0	Berglund's sm&bb7Pp	Christina Berglund	Order Administrator	Callahan	11						
10255	26	RIZSU	9	1996-07-12	1996-08-20	1996-07-15	3	14833	39	44	30	0	Richter Supermarkt	Michael Holz	Sales Manager	Dodsworth	4						
10267	27	FRANK	4	1996-07-20	1996-08-26	1996-08-06	1	20939	39	44	70	15	Frankenmeat	Peter Franken	Marketing Manager	Pearson	9						
10275	28	MAGAA	1	1996-08-07	1996-09-04	1996-08-09	1	2653	39	44	6	5	Maggiore Alimentari Runzi	Giovanni Rovelli	Marketing Manager	Davidis	9						
10280	29	BERUS	2	1996-08-14	1996-09-11	1996-08-12	1	866	55	192	20	0	Berglund's sm&bb7Pp	Christina Berglund	Order Administrator	Fulmer	4						

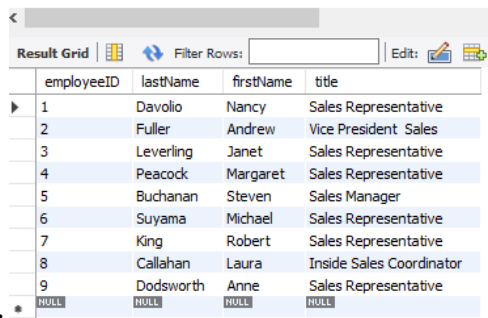
Screenshot:

Phase 2: Normalization and Denormalization

Following the import, normalization was performed to reduce redundancy and improve data integrity. Redundant data, particularly related to employee and product details, was identified and removed. Three key tables were designed: Employees, Products, and Orders. The Employees table stored employee information; the Products table contained product names and pricing details; and the Orders table linked products to employees with order-specific data such as quantity and date.

To efficiently populate these tables, the CREATE TABLE AS SELECT (CTAS) method was used. This approach allowed for the transformation of the original flat table structure into a set of relational tables aligned with normalization principles.

```
1 • SELECT * FROM employees;
```



The screenshot shows a database query result grid with the following data:

employeeID	lastName	firstName	title
1	Davolio	Nancy	Sales Representative
2	Fuller	Andrew	Vice President Sales
3	Leverling	Janet	Sales Representative
4	Peacock	Margaret	Sales Representative
5	Buchanan	Steven	Sales Manager
6	Suyama	Michael	Sales Representative
7	King	Robert	Sales Representative
8	Callahan	Laura	Inside Sales Coordinator
9	Dodsworth	Anne	Sales Representative
NULL	NULL	NULL	NULL

Screenshot:

```
1 • SELECT * FROM customers;
```

Result Grid				
Filter Rows:		Export:	Wrap Cell Content:	
customerID	companyName	contactName	contactTitle	
VINET	Vins et alcools Chevalier	Paul Henriot	Accounting Manager	
WARTH	Wartian Herkku	Pirkko Koskitalo	Accounting Manager	
ERNSH	Ernst Handel	Roland Mendel	Sales Manager	
RICSU	Richter Supermarkt	Michael Holz	Sales Manager	
FOLKO	Folk och f�r, HB	Maria Larsson	Owner	
GROSR	GROSELLA-Restaurante	Manuel Pereira	Owner	
RATTC	Rattlesnake Canyon Grocery	Paula Wilson	Assistant Sales Representative	
TOMSP	Toms Spezialit�ten	Karin Josephs	Marketing Manager	
BERGS	Berglunds snabbk�p	Christina Berglund	Order Administrator	
CHOPS	Chop-suey Chinese	Yang Wang	Owner	
MAGAA	Magazzini Alimentari Riuniti	Giovanni Rovelli	Marketing Manager	
HANAR	Hanari Carnes	Mario Pontes	Accounting Manager	
FRANK	Frankenversand	Peter Franken	Marketing Manager	
OTTIK	Ottilies K�f,seladen	Henriette Pfalzheim	Owner	
LILAS	LILA-Supermercado	Carlos Gonz�lez	Accounting Manager	
VICTE	Victualles en stock	Mary Saveley	Sales Agent	
BLONP	Blondesddsl p�tre et fils	Fr�d�rique Cit...	Marketing Manager	
LEHMS	Lehmanns Marktstand	Renate Messner	Sales Representative	
HILAA	HILARION-Abastos	Carlos Hern�ndez	Sales Representative	
SUPRD	Supr�mes d�licies	Pascale Cartrain	Accounting Manager	
CENTC	Centro comercial Moctezuma	Francisco Chang	Marketing Manager	
QUEDE	Que Del�cia	Bernardo Batista	Accounting Manager	
WHITC	White Clover Markets	Karl Jablonski	Owner	
MORGK	Morgenstern Gesundkost	Alexander Feuer	Marketing Assistant	
WELLI	Wellington Importadora	Paula Parente	Sales Manager	

1 • `SELECT * FROM suppliers;`

Result Grid				
Filter Rows:		Edit:		
Export/Import:		Wrap Cell Content:		
supplierID	companyName	contactName	contactTitle	
1	Exotic Liquids	Charlotte Cooper	Purchasing Manager	
2	New Orleans Cajun Delights	Shelley Burke	Order Administrator	
3	Grandma Kelly's Homestead	Regina Murphy	Sales Representative	
4	Tokyo Traders	Yoshi Nagase	Marketing Manager	
5	Cooperativa de Quesos 'Las Cabras'	Antonio del Valle Saavedra	Export Administrator	
6	Mayumi's	Mayumi Ohno	Marketing Representative	
7	Pavlova	Ltd	Ian Devling	
8	Specialty Biscuits	Ltd	Peter Wilson	
9	PB Knäckebröd AB	Lars Peterson	Sales Agent	
10	Refrescos Americanas LTDA	Carlos Diaz	Marketing Manager	
11	Heli Sushiken Waren GmbH & Co KG	Petra Winkler	Sales Manager	
12	Plutzer Lebensmittelgroßmärkte AG	Martin Bein	International Marketing Mgr	
13	Nord-Ost-Fisch Handelsgesellschaft...	Sven Petersen	Coordinator Foreign Mar...	
14	Formaggi Fortini srl	Elio Rossi	Sales Representative	
15	Norske Meierier	Beate Vileid	Marketing Manager	
16	Bigfoot Breweries	Cheryl Saylor	Regional Account Rep	
17	Svensk Sjöföda AB	Michael Björk	Sales Representative	
18	Aux joyeux ecclesiastiques	Guy L'Évêque Nodier	Sales Manager	
19	New England Seafood Cannery	Robb Merchant	Wholesale Account Agent	
20	Leka Trading	Chandra Leka	Owner	
21	Lyngbysild	Niels Petersen	Sales Manager	
22	Zaanse Snoepfabriek	Dirk Luchte	Accounting Manager	
23	Karkki Oy	Anne Heikkinen	Product Manager	
24	G'day	Mate	Wendy Mackenzie	
25	Ma Maison	Jean-Guy Lauzon	Marketing Manager	

1 • `SELECT * FROM Categories;`

Result Grid	
	 Filter Rows:
categoryID	categoryName
1	Beverages
2	Condiments
3	Confections
4	Dairy Products
5	Grains/Cereals
6	Meat/Poultry
7	Produce
8	Seafood
NULL	NULL

1 • `SELECT * FROM Products;`

Result Grid							
Filter Rows:							
Edit:   							
Export/Import:  							
Wrap Cell Content: 							
productID	productName	supplierID	categoryID	quantityPerUnit	unitPrice	unitsInStock	unit
1	Chai	1	1	10 boxes x 20 bags	18	39	0
2	Chang	1	1	24 - 12 oz bottles	19	17	40
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10	13	70
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22	53	0
5	Chef Anton's Gumbo Mix	2	2	36 boxes	2135	0	0
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25	120	0
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs	30	15	0
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40	6	0
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs	97	29	0
10	Ikura	4	8	12 - 200 ml jars	31	31	0
11	Queso Cabrales	5	4	1 kg pkg	21	22	30
12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs	38	86	0
13	Konbu	6	8	2 kg box	6	24	0
14	Tofu	6	7	40 - 100 g pkgs	2325	35	0
15	Genen Shouyu	6	2	24 - 250 ml bottles	155	39	0
16	Pavlova	7	3	32 - 500 g boxes	1745	29	0
17	Alice Mutton	7	6	20 - 1 kg tins	39	0	0
18	Carnarvon Tigers	7	8	16 kg pkg	625	42	0
19	Teatime Chocolate Biscuits	8	3	10 boxes x 12 pieces	92	25	0
20	Sir Rodney's Marmalade	8	3	30 gift boxes	81	40	0
21	Sir Rodney's Scones	8	3	24 pkgs x 4 pieces	10	3	40
22	Gustaf's Knäckebröd	9	5	24 - 500 g pkgs	21	104	0
23	Tunnbröd	9	5	12 - 250 g pkgs	9	61	0
24	Guaraná Fantástica	10	1	12 - 355 ml cans	45	20	0

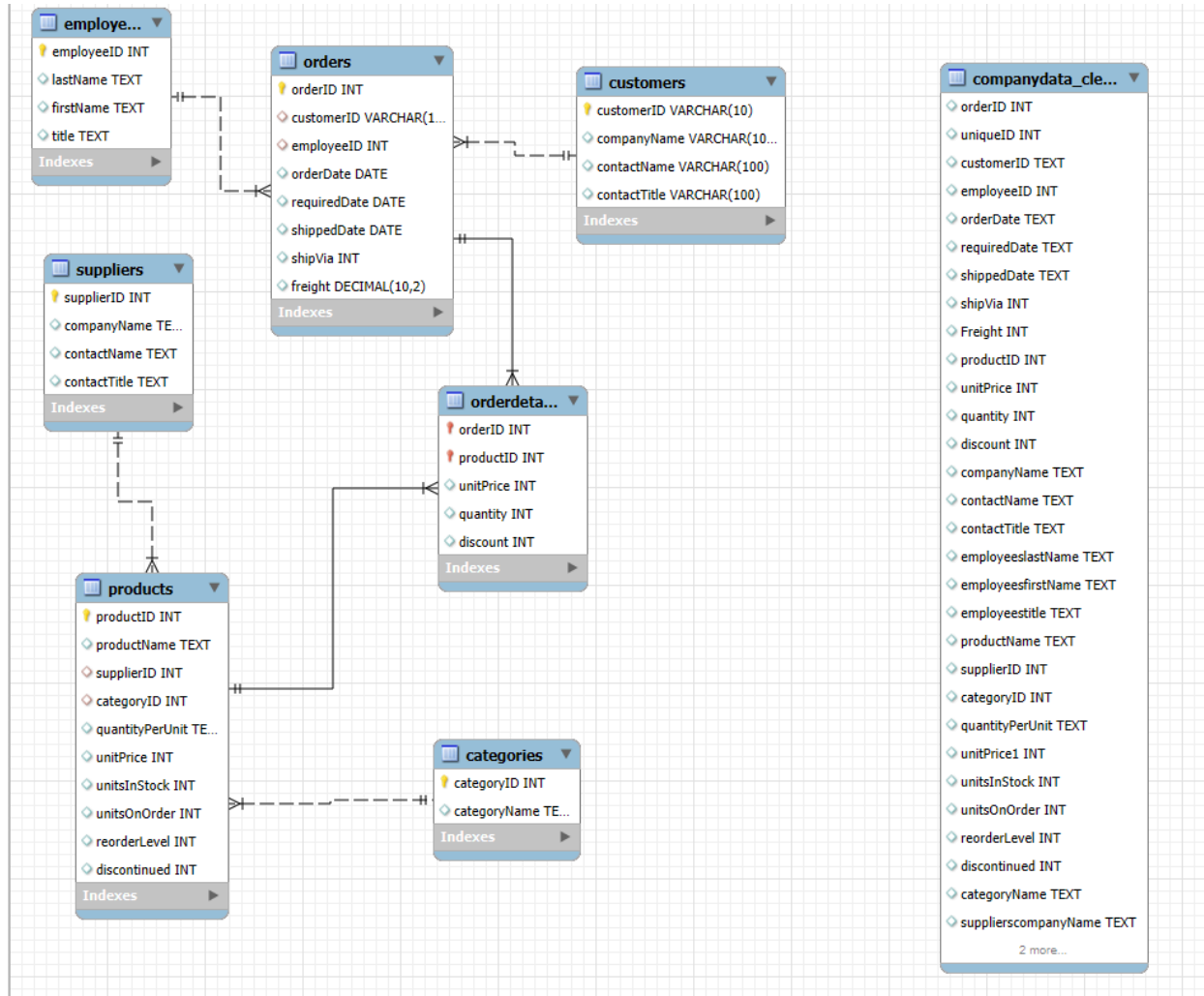
```
1 • SELECT * FROM Orders;
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:								
	orderID	customerID	employeeID	orderDate	requiredDate	shippedDate	shipVia	freight
▶	10248	VINET	5	1996-07-04	1996-08-01	1996-07-16	3	3238.00
	10249	TOMSP	6	1996-07-05	1996-08-16	1996-07-10	1	1161.00
	10250	HANAR	4	1996-07-08	1996-08-05	1996-07-12	2	6583.00
	10251	VICTE	3	1996-07-08	1996-08-05	1996-07-15	1	4134.00
	10252	SUPRD	4	1996-07-09	1996-08-06	1996-07-11	2	513.00
	10253	HANAR	3	1996-07-10	1996-07-24	1996-07-16	2	5817.00
	10254	CHOPS	5	1996-07-11	1996-08-08	1996-07-23	2	2298.00
	10255	RICSU	9	1996-07-12	1996-08-09	1996-07-15	3	14833.00
	10256	WELLI	3	1996-07-15	1996-08-12	1996-07-17	2	1397.00
	10257	HILAA	4	1996-07-16	1996-08-13	1996-07-22	3	8191.00
	10258	ERNSH	1	1996-07-17	1996-08-14	1996-07-23	1	14051.00
	10259	CENTC	4	1996-07-18	1996-08-15	1996-07-25	3	325.00
	10260	OTTIK	4	1996-07-19	1996-08-16	1996-07-29	1	5509.00
	10261	QUEDE	4	1996-07-19	1996-08-16	1996-07-30	2	305.00
	10262	RATTC	8	1996-07-22	1996-08-19	1996-07-25	3	4829.00
	10263	ERNSH	9	1996-07-23	1996-08-20	1996-07-31	3	14606.00
	10264	FOLKO	6	1996-07-24	1996-08-21	1996-08-23	3	367.00
	10265	BLONP	2	1996-07-25	1996-08-22	1996-08-12	1	5528.00
	10266	WARTH	3	1996-07-26	1996-09-06	1996-07-31	3	2573.00
	10267	FRANK	4	1996-07-29	1996-08-26	1996-08-06	1	20858.00
	10268	GROSR	8	1996-07-30	1996-08-27	1996-08-02	3	6629.00
	10269	WHITC	5	1996-07-31	1996-08-14	1996-08-09	1	456.00
	10270	WARTH	1	1996-08-01	1996-08-29	1996-08-02	1	13654.00
	10271	SPLIR	6	1996-08-01	1996-08-29	1996-08-30	2	454.00
	10272	RATTC	6	1996-08-02	1996-08-30	1996-08-06	2	9803.00
	10273	FRANK	4	1996-08-03	1996-08-31	1996-08-10	1	20858.00

Phase 3: Database Diagram Design (ER)

To visualize the logical structure of the database, an Entity-Relationship (ER) diagram was generated using MySQL Workbench's Reverse Engineer feature. This diagram illustrated the relationships among the Employees, Products, and Orders tables. The diagram helped validate the one-to-many relationships, such as one employee handling multiple orders and each product appearing in multiple orders.

Screenshot:



Phase 4: Table Alterations

Once the ER diagram was finalized, structural adjustments were made to the tables to align with the defined relationships. Primary keys were added to uniquely identify records in each table, and foreign key constraints were introduced to enforce referential integrity between related entities. Additionally, NOT NULL constraints were applied where appropriate, and data types were checked and adjusted to ensure consistency across the schema. These modifications were made using the ALTER TABLE command in MySQL Workbench.

Screenshot:

The screenshot displays the MySQL Workbench interface. The top pane shows a series of SQL queries for creating and populating tables. The bottom pane shows the 'Output' window with a table of execution results.

SQL Queries:

```
1 DROP TABLE IF EXISTS OrderDetails, Orders, Products, Categories, Suppliers, Customers, Employees;
2
3 -- 1. Employees
4 CREATE TABLE Employees (
5     employeeID INT PRIMARY KEY,
6     lastName VARCHAR(50),
7     firstName VARCHAR(50),
8     title VARCHAR(100)
9 );
10
11 INSERT INTO Employees
12 SELECT DISTINCT
13     employeeID,
14     employeeslastName,
15     employeesfirstName,
16     employeeestitle
17 FROM companydata_clean;
18
19 -- 2. Customers
20 CREATE TABLE Customers (
21     customerID VARCHAR(10) PRIMARY KEY,
22     companyName VARCHAR(100),
```

Output Window:

#	Time	Action	Message
105	15:30:36	CREATE TABLE Employees (employeeID INT PRIMARY KEY, lastName VARCHAR(50), firstName VARCHAR(50), title VARCHAR(100))	0 row(s) affected
106	15:30:36	INSERT INTO Employees SELECT DISTINCT employeeID, employeeslastName, employeesfirstName, employeeestitle FROM companydata_clean	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
107	15:30:36	CREATE TABLE Customers (customerID VARCHAR(10) PRIMARY KEY, companyName VARCHAR(100), contactName VARCHAR(100), c...	0 row(s) affected
108	15:30:36	INSERT INTO Customers SELECT DISTINCT customerID, companyName, contactName, contactTitle FROM companydata_clean	85 row(s) affected Records: 85 Duplicates: 0 Warnings: 0
109	15:30:36	CREATE TABLE Suppliers (supplierID INT PRIMARY KEY, companyName VARCHAR(100), contactName VARCHAR(100), contactTitle VA...	0 row(s) affected
110	15:30:36	INSERT INTO Suppliers SELECT DISTINCT supplierID, supplierscompanyName, supplierscontactName, supplierscontactTitle FROM compa...	29 row(s) affected Records: 29 Duplicates: 0 Warnings: 0
111	15:30:36	CREATE TABLE Categories (categoryID INT PRIMARY KEY, categoryName VARCHAR(100))	0 row(s) affected
112	15:30:36	INSERT INTO Categories SELECT DISTINCT categoryID, categoryName FROM companydata_clean	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
113	15:30:36	CREATE TABLE Products (productID INT PRIMARY KEY, productName VARCHAR(100), supplierID INT, categoryID INT, quantityPerUn...	0 row(s) affected
114	15:30:36	INSERT INTO Products SELECT DISTINCT productID, productName, supplierID, categoryID, quantityPerUnit, unitPrice1, unitsInSto...	77 row(s) affected Records: 77 Duplicates: 0 Warnings: 0
115	15:30:36	CREATE TABLE Orders (orderID INT PRIMARY KEY, customerID VARCHAR(10), employeeID INT, orderDate DATE, requiredDate DAT...	0 row(s) affected
116	15:30:36	INSERT INTO Orders SELECT DISTINCT orderID, customerID, employeeID, orderDate, requiredDate, shippedDate, shipVia, Freig...	378 row(s) affected Records: 378 Duplicates: 0 Warnings: 0
117	15:30:36	CREATE TABLE OrderDetails (orderID INT, productID INT, unitPrice DECIMAL(10,2), quantity INT DEFAULT 1 CHECK (quantity >= 0), di...	0 row(s) affected
118	15:30:36	INSERT INTO OrderDetails SELECT orderID, productID, unitPrice, quantity, discount FROM companydata_clean	1000 row(s) affected Records: 1000 Duplicates: 0 Warnings: 0

Phase 5: Views, Triggers, and Stored Procedures

To enhance the functionality of the database, several dynamic elements were implemented. A view named EmployeeSales was created to summarize the total quantity sold and total revenue generated by each employee. A trigger titled update_stock was developed to automatically decrease a product's stock level whenever a new order was inserted into the Orders table. Furthermore, a stored procedure called CheckRestock was written to retrieve a list of products with low stock levels, allowing for proactive inventory management.

Screenshot:

```
1  -- View: quantity sold and revenue per employee
2  • CREATE OR REPLACE VIEW EmployeeSales AS
3  SELECT
4      e.employeeID,
5      e.firstName,
6      e.lastName,
7      SUM(od.quantity) AS totalQuantitySold,
8      SUM(od.quantity * od.unitPrice * (1 - (od.discount / 100))) AS totalRe
9  FROM Employees e
10 JOIN Orders o ON e.employeeID = o.employeeID
11 JOIN OrderDetails od ON o.orderID = od.orderID
12 GROUP BY e.employeeID, e.firstName, e.lastName;
13
14
15 -- Reduce stock when order placed
16 • DROP TRIGGER IF EXISTS ReduceStockAfterOrder;
17
18 DELIMITER $$
19 • CREATE TRIGGER ReduceStockAfterOrder
20 AFTER INSERT ON OrderDetails
21 FOR EACH ROW
22 BEGIN
```

Result Grid				
Filter Rows:		Export:		Wrap Cell Content: IA
employeeID	firstName	lastName	totalQuantitySold	totalRevenue
1	Nancy	Davolio	3741	773626.65000000
2	Andrew	Fuller	2784	843469.81000000
3	Janet	Leverling	3421	993709.81000000
4	Margaret	Peacock	5301	1376082.45000000
5	Steven	Buchanan	1549	252763.25000000
6	Michael	Suyama	1745	951776.92000000
7	Robert	King	1881	488770.29000000
8	Laura	Callahan	2639	569390.63000000
9	Anne	Dodsworth	1011	249747.46000000

```

30 • -- Check for products low in stock
31 DROP PROCEDURE IF EXISTS CheckReorderProducts;
32
33 DELIMITER $$
34 • CREATE PROCEDURE CheckReorderProducts()
35 BEGIN
36     SELECT
37         productID,
38         productName,
39         unitsInStock,
40         reorderLevel
41     FROM Products
42     WHERE unitsInStock <= reorderLevel;
43 END$$
44 DELIMITER ;

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Cont

productID	productName	unitsInStock	reorderLevel
2	Chang	17	25
3	Aniseed Syrup	13	25
5	Chef Anton's Gumbo Mix	0	0
11	Queso Cabrales	22	30
17	Alice Mutton	0	0
21	Sir Rodney's Scones	3	5
29	Thüringer Rostbratwurst	0	0
30	Nord-Ost Matjeshering	10	15
31	Gorgonzola Telino	0	20
32	Mascarpone Fabioli	9	25
37	Gravad lax	11	25
43	Ipoh Coffee	17	25
45	Rogede sild	5	15
48	Chocolade	15	25
49	Maxilaku	10	15
53	Perth Pasties	0	0
56	Gnocchi di nonna Alice	21	30
64	Wimmers gute Semmelkn...	22	30
66	Louisiana Hot Spiced Okra	4	20
68	Scottish Longbreads	6	15

Phase 6: User Management and Access Control

Database access was structured by creating two users with different levels of permissions. A user named AlexSmith was granted full DBA (Database Administrator) privileges, allowing for complete control over the database. Another user, JamieLee, was assigned read-only access, suitable for data analysis or reporting roles. Permissions were configured using the GRANT command, ensuring a secure and role-appropriate access model.

Screenshot:

```
2 • DROP USER IF EXISTS 'AlexSmith'@'localhost';
3 • DROP USER IF EXISTS 'JamieLee'@'localhost';
4
5 • CREATE USER 'AlexSmith'@'localhost' IDENTIFIED BY 'meowmeowmeowmeowMEOW';
6 • GRANT ALL PRIVILEGES ON companydb.* TO 'AlexSmith'@'localhost';
7
8   -- Create the analyst user
9 • CREATE USER 'JamieLee'@'localhost' IDENTIFIED BY 'ILoveDataBaseSystem!';
10 • GRANT SELECT ON companydb.* TO 'JamieLee'@'localhost';
11
12 • FLUSH PRIVILEGES;
13
14 • SHOW GRANTS FOR 'AlexSmith'@'localhost';
15 • SHOW GRANTS FOR 'JamieLee'@'localhost';
```

result Grid	Filter Rows:	Export: 	Wrap Cell Content: 
Grants for JamieLee@localhost			
GRANT USAGE ON *.* TO 'JamieLee'@'localh...			
GRANT SELECT ON `companydb`,* TO 'Jamiel...			

Phase 7: Database Backup

The final stage of the project focused on data preservation and disaster recovery. MySQL Workbench's Data Export feature was used to generate a full backup of both the database schema and data. The resulting .sql file was then uploaded to a secure cloud storage service, such as a hard drive, ensuring that the database could be restored in the event of data loss or corruption.

Screenshot:

Name	Date modified	Type	Size
CompanyData_clean	5/6/2025 6:42 PM	Microsoft Excel C...	265 KB
companydb_backup	5/8/2025 4:46 PM	SQL Text File	378 KB
DBSProj	5/8/2025 3:28 PM	SQL Text File	3 KB
Phase1	5/8/2025 3:05 PM	SQL Text File	1 KB
Phase2	5/8/2025 3:06 PM	SQL Text File	1 KB
Phase3	5/8/2025 3:22 PM	MySQL Workbenc...	53 KB
Phase4	5/8/2025 3:32 PM	SQL Text File	3 KB
Phase5	5/8/2025 4:00 PM	SQL Text File	2 KB
Phase6	5/8/2025 4:39 PM	SQL Text File	1 KB

Team Contributions:

All of the team members contributed to the success of this project through distinct responsibilities, as outlined below.

Emma Bahr was primarily responsible for the data cleaning process, including reformatting date fields and standardizing numerical values. She also led the effort in setting up user accounts and defining permission levels. Kyle Gibson focused on database creation and schema design, ensuring the normalization of data and building the ER diagram. He also developed the necessary SQL scripts, including views, triggers, and stored procedures. The final report and packaging of deliverables were a collaborative effort between both team members.